# SHAHAB ALAM

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• LeetCode | • GitHub | • Portfolio

## SUMMARY

Full-Stack Agentic AI Developer skilled in Python, LangChain, LangGraph, RAG, and LLMs, with hands-on experience building AI-powered applications using React, FastAPI, Node.js, Docker, and AWS. Strong in designing agentic workflows and solving complex problems, with 300+ DSA problems solved.

## SKILLS

**Languages:** Python, JavaScript, C, C++, HTML, CSS
**Frontend:** React, React Native, Tailwind CSS, Framer Motion, Angular (Basics), RxJS (Basics)
**Backend:** Node.js, Express.js, FastAPI, JWT, REST APIs, Microservices, Hibernate (Basics)
**Gen & Agentic AI:** Generative AI, LLMs, LangChain, LangGraph, MCP, LCEL, Runnables, Sequential Workflows, Parallel (Fan-Out/Fan-In), Conditional Routing, Iterative (Cyclic) Workflows, Human-in-the-Loop (HITL), State Machines, Multi-Agent Systems, Tool Use, Agentic RAG, RAG, ChromaDB, Structured Outputs, Context Engineering, Prompt Engineering, LlamaIndex, Ollama, Hugging Face
**Database:** MongoDB, MySQL, PostgreSQL, OpenSearch (Elasticsearch), MinIO (S3-compatible), Oracle (Basics)
**DevOps & Cloud:** AWS (EKS, S3, Lambda, SQS), Docker, Kubernetes, CI/CD, GitHub Actions , Linux (Basics)
**Tools & Core:** Git, GitHub, Agile/Scrum, DSA, OOPs, Microservices Architecture, Orchestration, Distributed Systems, Redis (Basics), DBMS, OS

## PROJECTS

**NyayaAI – Legal Intelligence System** | *MERN, LangChain, LangGraph, Gemini API, RAG, LCEL, WebAssembly, Socket.io* | Live

- Architected a unified 3-in-1 legal SaaS platform – integrating AI document intelligence, an algorithmic lawyer matchmaking marketplace, and a 15-module enterprise practice management workspace
- Built an advanced RAG pipeline using LangChain LCEL with Markdown-aware recursive chunking, semantic embeddings, and Pydantic-constrained structured outputs to power a deterministic "Silence Detector" – proactively flagging asymmetrical or predatory legal clauses pre-signature with schema-bound JSON responses
- Implemented an **Agentic AI layer** using a LangGraph conditional workflow – legal queries are classified and routed through specialized tool-use nodes; few-shot system prompts enforce hallucination-resistant outputs, and retrieved content is treated strictly as *data/evidence* for built-in prompt-injection protection
- Developed a zero-trust "Private Mode" with PDF.js and WebAssembly (WASM) to parse legal documents strictly in the client's browser memory, ensuring DPDP Act 2023 compliance with zero server-side data exposure; built real-time collaboration via Socket.io and cross-tenant isolation using JWT-based RBAC on a Node.js/Express backend

**CampusQuery – STCET AI Assistant** | *Python, LangChain, LangGraph, RAG, Embeddings, Web Scraping*

- Built a production-grade RAG knowledge assistant for STCET by crawling and mapping the college website – 677 URLs discovered → 659 filtered → 309 pages selected → 310 structured Markdown documents covering Admissions, Academics, Fees, Faculty, Placements, and College Info
- Designed and benchmarked 4 chunking strategies against a custom **56-question gold-answer benchmark**: Fixed (63.5%) → Semantic (71.4%) → Recursive + Overlap (78.6%) → **Markdown-aware + Recursive + Overlap (89.3%)** – improving retrieval hit rate by **25.8 percentage points**
- Implemented a LangGraph **conditional workflow** for prompt-injection protection – retrieved content is treated strictly as *data/evidence*, not instructions; suspicious inputs are routed through a separate conditional path rather than the normal RAG flow; validated with custom injection test cases

## CERTIFICATIONS

- Agentic AI Full Course – Sheryians AI School (LangGraph Workflows, HITL, Multi-Agent Systems)
- Generative AI with LLMs – DeepLearning.AI / Coursera
- Full Stack MERN Development – Apna College
- Meta Front-End Developer Professional Certificate – Coursera
- HackerRank SQL (Advanced) Certification

## ACHIEVEMENTS

- Deployed NyayaAI to live production on AWS with Docker, GitHub Actions CI/CD, and zero-downtime release strategy – serving real users at scale
- Solved 300+ DSA problems on LeetCode demonstrating strong analytical thinking and systematic problem-solving ability
- Built 5 Agentic AI workflow systems (Sequential, Parallel, Conditional, Iterative, Human-in-the-Loop) and 6 GenAI projects covering RAG, LCEL, Runnables, Structured Outputs, and Tool Use – all from scratch

## EDUCATION

**B.Tech in Computer Science Engineering** **Aug 2022 – July 2026**
*St Thomas College of Engineering and Technology, Kolkata*